

Liquidation-Aware Market Making in Perpetual Futures: Optimal Quoting under a Self-Exciting Forced-Liquidation Process, and an Empirical Test of Its Premise

Boon Chuan Lim

Independent Researcher, Singapore

boonchuan@singapore.to

ORCID: 0009-0005-8477-9393

This version: 29 May 2026

Abstract

A market maker on a perpetual futures venue faces forced liquidations: order flow that is exogenous, directional, and self-exciting. I model signed liquidation flow as a bivariate marked Hawkes process and embed its conditional intensity in an Avellaneda–Stoikov problem. A maker left with inventory must unwind it into a book consumed by same-side forced flow; I derive this exit cost rather than assume it, and show it is quadratic in inventory and linear in the matched intensity. That cost breaks the inventory symmetry of the value function and yields a clean prediction: if liquidation flow is toxic, the optimal reservation price leans away from the more active cascade direction, linearly in the intensity gap. I then test the premise on Hyperliquid, using a complete on-chain liquidation series that collapses roughly 42,000 raw forced-liquidation fills into several thousand cascade events per contract. The liquidation process is subcritical and self-exciting, with spectral radius near 0.81 and a decay of tens of seconds. But the toxicity premise fails on the most liquid contract. Passive fills taken during elevated same-side liquidation intensity are not adversely selected; they earn favourable forward mark movement, consistent with cascades that mean-revert over about a minute. Resting depth tilts toward, not away from, the active cascade, and there is no liquidation-specific spread premium. On this contract the maker should lean into liquidations, not away. The within-venue picture across contracts is heterogeneous, and I report it honestly: the result is cleanest on Bitcoin, underpowered on Ether, and mixed on Solana, a pattern that motivates a pre-registered hypothesis linking liquidation toxicity to book depth. The model is internally correct; its empirical premise does not hold where a deep, fast liquidation backstop makes cascades revert. The contribution is a tractable model whose scope this delimits, a reusable on-chain liquidation dataset, and direct evidence on when liquidation-aware quoting should and should not matter.

Keywords: market making; perpetual futures; Hawkes processes; forced liquidation; mean reversion; adverse selection.

JEL classification: G12; G13; G14; C61.

1. Introduction

A market maker quotes a bid and an ask, earns the spread, and carries inventory risk. The canonical treatment is Avellaneda and Stoikov (2008), building on Ho and Stoll (1981). Later work relaxes the Poisson fill assumption; Jusselin (2021) replaces it with general Hawkes order flow. These models assume the flow that fills the maker is, in aggregate, directionally balanced, and that the cost of carrying inventory is symmetric in its sign.

Perpetual futures appear to break both assumptions. A perpetual contract has no expiry, ties its price to the underlying through funding, and closes under-margined positions by force. A forced liquidation is mechanical, one-directional, and pushes the price further in the direction that triggered it; a forced sale lowers the price, breaching the margin of other longs, triggering more forced sales. Liquidations cluster, and the clusters are signed. The natural conjecture, and the one this paper formalises, is that this flow is the dominant source of adverse selection for a passive maker: the resting bid is filled by exactly the forced sells that precede further falls, so the maker should quote away from the active cascade.

I build the model this conjecture implies. Signed liquidation flow is a bivariate marked Hawkes process. Its conditional intensity enters an Avellaneda–Stoikov problem as a state variable. The maker who is left with inventory unwinds it into a book consumed by same-side forced flow, and I derive the resulting exit cost rather than assume it: it is quadratic in inventory and linear in the matched liquidation intensity, following Almgren and Chriss (2001). That cost breaks the inventory symmetry of the value function. The result, Proposition 1, is sharp: when the long- and short-liquidation intensities differ, the optimal reservation price shifts away from the more active direction, linearly in the intensity gap, even at zero inventory. The half-spread carries no such first-order asymmetry. Classical and persistent-order-flow market making are recovered in the no-liquidation limit.

The prediction has a premise: that liquidation flow is toxic. The second half of the paper tests it. I assemble a complete, on-chain liquidation series for Hyperliquid from the trade- and fill-level data, independent of how each liquidation is routed, and collapse the mechanically clustered fills into cascade events — 1,087 long and 2,233 short on Bitcoin over the sample month, with comparable constructions on Ether and Solana. The cascade process is exactly as the model supposes in its dynamics: subcritical, with spectral radius near 0.81, self-exciting, directional, decaying over tens of seconds. The premise, however, does not hold on the most liquid contract. A passive fill taken while same-side liquidation intensity is high is not adversely selected; it earns favourable forward mark movement, and the gain accrues over the first minute as the cascade reverts. Aggregate resting depth tilts toward the active cascade, not away. There is no liquidation-specific spread premium beyond volatility. The three findings are mutually consistent: on this contract a cascade is an overshoot that mean-reverts, so providing liquidity into it is rewarded. The maker should lean toward the cascade, the opposite of what the model prescribes.

I report this directly rather than search for a specification that confirms the model. The model is not wrong; its premise is venue-specific. Liquidation-aware skewing is the right policy where forced liquidations carry information or impose lasting impact, so that fills against them are genuinely toxic, which describes venues with thin backstop liquidity, slow liquidation engines, or large positions relative to depth. Hyperliquid, with a deep protocol backstop and fast partial liquidations, is not such a venue over the sample; its cascades are absorbed and reverse.

The contribution is three-fold. First, a tractable market-making model in which a microfounded, intensity-dependent exit cost makes the optimal reservation price respond to directional liquidation pressure, with the response characterised in closed form. Second, a reusable method and dataset: a routing-independent, on-chain liquidation series with a robust cascade construction, and a bivariate Hawkes characterisation of the cascade process. Third, direct evidence on the premise that liquidation-aware quoting depends on, showing it fails on a major venue because cascades mean-revert, which bounds where the quoting result should be expected to bite and motivates testing it where the premise plausibly holds. The closest prior work is the market-making lineage from Avellaneda and Stoikov (2008) through Guéant, Lehalle and

Fernandez-Tapia (2013) and Cartea, Jaimungal and Penalva (2015); the Hawkes-in-finance literature surveyed by Bacry, Mastromatteo and Muzy (2015) and the persistent-order-flow model of Jusselin (2021); and the liquidation-clustering evidence of Cao and Gang (2026). The cell this paper occupies, joining a microfounded liquidation exit cost to a quoting rule and then testing the toxicity premise on-chain, was empty.

2. The market

2.1. Mark price and the maker

Fix a filtered probability space and a finite horizon T . The perpetual mark price S follows

$$dS_t = \sigma dW_t + \eta (dN_t^- - dN_t^+), \quad (1)$$

where W is a standard Brownian motion, $\sigma > 0$ is diffusive volatility, and N^+ , N^- are the counting processes for long and short forced liquidations. A long liquidation is a forced sale and lowers the price; a short liquidation is a forced purchase and raises it. The constant $\eta > 0$ is the average price impact of one liquidation. The sign convention encodes the mechanical feedback: a long liquidation lowers S , which is what makes the next long liquidation more likely.

The maker posts a bid at $S - \delta^b$ and an ask at $S + \delta^a$, with non-negative offsets δ^b , δ^a it controls. Own fills arrive as Poisson processes with intensities declining in the offset,

$$\lambda^b(\delta^b) = A e^{-(k \delta^b)}, \quad \lambda^a(\delta^a) = A e^{-(k \delta^a)}, \quad (2)$$

the exponential fill of Avellaneda and Stoikov (2008), with A , $k > 0$. A bid fill raises inventory q by one; an ask fill lowers it. Cash is X .

2.2. The liquidation process

The pair (N^+, N^-) of long- and short-liquidation counting processes is a bivariate Hawkes process. I use the normalised exponential kernel, so that the conditional intensities are

$$\lambda_t^+ = \mu^+ + \sum_{\text{over } t_k^+ < t} \text{of } a_{++} \beta e^{(-\beta(t-t_k^+))} + \sum_{\text{over } t_k^- < t} \text{of } a_{+-} \beta e^{(-\beta(t-t_k^-))}, \quad (3a)$$

$$\lambda_t^- = \mu^- + \sum_{\text{over } t_k^- < t} \text{of } a_{--} \beta e^{(-\beta(t-t_k^-))} + \sum_{\text{over } t_k^+ < t} \text{of } a_{-+} \beta e^{(-\beta(t-t_k^+))}, \quad (3b)$$

with baseline rates μ^+ , $\mu^- > 0$ and common decay $\beta > 0$. A single decay β is maintained across all four excitation channels to preserve parameter parsimony and stability over thinner data slices, and to keep the branching matrix directly comparable across specifications; the rolling fits below show β is in any case stable across windows. Because $\beta e^{(-\beta s)}$ integrates to one over $s > 0$, the coefficient a_{ij} is exactly the L^1 -norm of the $j \rightarrow i$ excitation kernel, that is the expected number of direct type- i offspring generated by a single type- j event. The branching matrix is therefore $\Gamma = [[a_{++}, a_{+-}], [a_{-+}, a_{--}]]$ itself, with no further normalisation, and the process is asymptotically stationary when its spectral radius satisfies $\rho(\Gamma) < 1$ (Hawkes 1971; Bacry, Mastromatteo and Muzy 2015). The pair (λ^+, λ^-) is Markov, driven by the jumps of (N^+, N^-) . I write the kernel in this normalised form precisely so that the estimated amplitudes are branching ratios and $\rho(\Gamma)$ is read directly off the amplitude matrix; an unnormalised kernel would require dividing by β to recover the same quantities.

2.3. The exit cost of inventory

This is the step that distinguishes the perpetual setting. Suppose the maker holds inventory q and must return to flat by trading it out through marketable orders. Following Almgren and Chriss (2001), splitting the position into a schedule of child orders and summing temporary impact gives an execution cost quadratic in size. On a perpetual venue the per-unit temporary impact is not constant: the maker unwinds into a book already being consumed by same-side forced flow, so each child order pays more when the matched liquidation intensity is higher. Writing the per-unit impact as $\kappa_0(1 + \chi \lambda_{\text{dir}})$ for a small constant $\chi > 0$, where λ_{dir} is λ^+ if $q > 0$ (a long unwound by selling) and λ^- if $q < 0$, the expected exit cost is

$$C(q, \lambda^+, \lambda^-) = \frac{1}{2} \kappa_0 (1 + \chi \lambda_{\text{dir}}) q^2, \quad (4)$$

quadratic in inventory and linear in the matched intensity. The dependence on the sign of q through λ_{dir} is the only place where the two liquidation directions enter the maker's problem asymmetrically. Everything in the next two sections follows from it. Note that C is convex in q and that its curvature, not merely its level, depends on the intensity state; this is what a linear mark-to-market term alone would miss.

3. The control problem

The maker maximises expected terminal wealth net of the exit cost it would pay to flatten, under a running inventory penalty. Mark-to-market wealth is $V_t = X_t + q_t S_t$. The value function is

$$u(t, x, q, s, \lambda^+, \lambda^-) = \sup E[X_T + q_T S_T - C(q_T, \lambda_T^+, \lambda_T^-) - \varphi \int_t^T q_s^2 ds], \quad (5)$$

the supremum over admissible δ^b, δ^a , with $\varphi \geq 0$ the running inventory aversion and C the exit cost (4) charged on the terminal position. Because (λ^+, λ^-) is Markov, dynamic programming applies on a finite-dimensional state.

Let $\Delta^+ u$ and $\Delta^- u$ denote the change in u across a long and a short liquidation, combining the price jump in (1) and the intensity jump implied by (3). The Hamilton–Jacobi–Bellman equation is

$$0 = \partial_t u + \frac{1}{2} \sigma^2 \partial_{ss} u - \varphi q^2 + \lambda^+ \Delta^+ u + \lambda^- \Delta^- u + H^b + H^a, \quad (6)$$

with control Hamiltonians $H^b = \sup \text{over } \delta^b \text{ of } A e^{(-k\delta^b)} [u(\cdot, x - (s - \delta^b), q + 1, \cdot) - u]$ and $H^a = \sup \text{over } \delta^a \text{ of } A e^{(-k\delta^a)} [u(\cdot, x + (s + \delta^a), q - 1, \cdot) - u]$.

Use the ansatz $u = x + q s + \theta(t, q, \lambda^+, \lambda^-)$. The cash and mark-to-market terms separate; θ carries inventory and liquidation risk and inherits the terminal condition $\theta(T, q, \lambda^+, \lambda^-) = -C(q, \lambda^+, \lambda^-)$. The first-order conditions give the optimal offsets

$$\delta^{b*} = 1/k - [\theta(t, q+1, \lambda^+, \lambda^-) - \theta(t, q, \lambda^+, \lambda^-)], \quad (7a)$$

$$\delta^{a*} = 1/k + [\theta(t, q-1, \lambda^+, \lambda^-) - \theta(t, q, \lambda^+, \lambda^-)]. \quad (7b)$$

Two observables summarise the quotes. The half-spread and the reservation shift are $\psi = \frac{1}{2}(\delta^{b*} + \delta^{a*})$ and $r = \frac{1}{2}(\delta^{a*} - \delta^{b*})$. The half-spread ψ measures how wide the maker quotes; the reservation shift r measures how far the quote pair is centred from the mark, with $r > 0$ a centring above the mark and $r < 0$ below. From (7),

$$\psi = 1/k - \frac{1}{2} [\theta(q+1) + \theta(q-1) - 2\theta(q)], \quad r = -\frac{1}{2} [\theta(q+1) - \theta(q-1)], \quad (8)$$

so the spread is governed by the discrete curvature of θ in inventory and the reservation shift by its discrete gradient. This decomposition is the key to the next section: a symmetric model makes θ even in q , which

forces $r = 0$ at $q = 0$ while leaving ψ positive. The liquidation asymmetry acts through the gradient, hence through r alone.

4. The asymmetric reservation price

I work at $q = 0$, the cleanest case, where any asymmetry cannot be attributed to held inventory. Two results follow: a lemma that the exit cost (4) makes θ non-even in q exactly when the intensities differ, and a closed form for the reservation shift under the quadratic expansion of Guéant, Lehalle and Fernandez-Tapia (2013).

4.1. Non-evenness of the value function

Lemma 1. Let θ solve (6) with terminal condition $\theta(T, q, \lambda^+, \lambda^-) = -C(q, \lambda^+, \lambda^-)$ and exit cost (4). Suppose the structure is symmetric, $a_{++} = a_{--}$, $a_{+-} = a_{-+}$, $\mu^+ = \mu^-$. Then $\theta(t, q, \lambda^+, \lambda^-) = \theta(t, -q, \lambda^-, \lambda^+)$ for all t, q . Consequently θ is even in q if and only if $\lambda^+ = \lambda^-$; if $\lambda^+ \neq \lambda^-$ then $\theta(t, 1, \lambda^+, \lambda^-) \neq \theta(t, -1, \lambda^+, \lambda^-)$.

Proof. Consider the reflection P that maps $q \mapsto -q$ and swaps the two liquidation labels, $(\lambda^+, N^+, \mu^+) \leftrightarrow (\lambda^-, N^-, \mu^-)$. Under the symmetric structure the generator of $(q, \lambda^+, \lambda^-)$ is invariant under P : the price dynamics (1) are odd in the liquidation labels, the fill operators in (6) exchange bid and ask under $q \mapsto -q$, the running penalty ϕq^2 is even, and the exit cost satisfies $C(-q, \lambda^+, \lambda^-) = C(q, \lambda^-, \lambda^+)$ because λ_{dir} follows the sign of q . Since both the generator and the terminal condition are P -invariant, so is the value function: $\theta(t, q, \lambda^+, \lambda^-) = \theta(t, -q, \lambda^-, \lambda^+)$. Setting $q = 1$ gives $\theta(t, 1, \lambda^+, \lambda^-) = \theta(t, -1, \lambda^-, \lambda^+)$. The right-hand side equals $\theta(t, -1, \lambda^+, \lambda^-)$ only when the arguments λ^+ and λ^- may be exchanged without changing the value, which under strict monotonicity of C in λ_{dir} holds precisely when $\lambda^+ = \lambda^-$. Hence θ is even in q at the origin if and only if $\lambda^+ = \lambda^-$. ■

Lemma 1 isolates the mechanism. It is not the Hawkes dynamics that break the symmetry, nor the price impact, both of which are present in symmetric form. It is the direction-matched exit cost: long inventory is costly in proportion to λ^+ and short inventory in proportion to λ^- , so when the two intensities differ the maker values a long and a short unit differently even from a flat book.

4.2. Closed form for the reservation shift

To make the shift explicit, expand θ to second order in inventory, $\theta(t, q, \lambda^+, \lambda^-) \approx \theta_0(t) + g(t, \lambda^+, \lambda^-) q - \frac{1}{2} \Gamma_0(t, \lambda^+, \lambda^-) q^2$, the form used by Guéant, Lehalle and Fernandez-Tapia (2013). The quadratic coefficient Γ_0 collects the running penalty and the curvature of the exit cost; the linear coefficient g collects the gradient. The terminal condition fixes $g(T, \lambda^+, \lambda^-)$ proportional to $\lambda^+ - \lambda^-$. Substituting into (8) at $q = 0$,

$$r(t) = -g(t, \lambda^+, \lambda^-) = -b(t) \cdot (\lambda^+ - \lambda^-), \quad (9)$$

for a coefficient $b(t) > 0$ that depends on κ_0, χ , the fill parameters A, k , and the horizon, and that is constant in the stationary long-horizon limit. The reservation shift is linear in the intensity gap with a strictly negative slope: when $\lambda^+ > \lambda^-$ the maker centres its quotes below the mark, leaning away from the side bearing the heavier long-liquidation flow.

Proposition 1 (reservation shift against the cascade). Under (1)–(5) with exit cost (4), the optimal flat-inventory reservation shift satisfies $r = -b \cdot (\lambda^+ - \lambda^-)$ with $b > 0$. Hence $r = 0$ when $\lambda^+ = \lambda^-$; $r < 0$ when $\lambda^+ > \lambda^-$; $r > 0$ when $\lambda^+ < \lambda^-$; and $\partial r / \partial (\lambda^+ - \lambda^-) < 0$. The half-spread ψ is even in the gap and carries no first-order asymmetry.

Proof. By Lemma 1, under symmetric structure $\theta(t, 1, \lambda^+, \lambda^-) - \theta(t, -1, \lambda^+, \lambda^-)$ vanishes at $\lambda^+ = \lambda^-$ and is otherwise non-zero, so by (8) $r = 0$ there and only there. The closed form (9) gives the sign and slope: $b > 0$ because $\kappa_0, \chi > 0$ and the fill kernel is decreasing, so r decreases in the gap. For the spread, (8) shows ψ depends on $\theta(q+1) + \theta(q-1) - 2\theta(q)$, which by the reflection identity of Lemma 1 is invariant under $\lambda^+ \leftrightarrow \lambda^-$ and therefore even in the gap; its first-order term in $\lambda^+ - \lambda^-$ is zero. ■

The proposition is the empirical content. A symmetric-flow model predicts $r = 0$ at flat inventory irrespective of recent liquidation activity. This model predicts a reservation shift linear in the contemporaneous intensity gap, with a sign that leans away from the active cascade. The numerical solution of (6) confirms (9): the flat-inventory shift is exactly linear in $\lambda^+ - \lambda^-$, vanishes at the origin, and reverses sign under relabelling, with no accompanying first-order change in the half-spread.

Corollary (nesting). As $\chi \rightarrow 0$ the exit cost loses its intensity dependence, g becomes even in q , and $r \equiv 0$: the quotes reduce to the symmetric Guéant, Lehalle and Fernandez-Tapia (2013) form, and with $\eta \rightarrow 0$ to the Avellaneda and Stoikov (2008) spread. Merging the two liquidation components into one undirected flow recovers the persistent-order-flow maker of Jusselin (2021). The model is a strict generalisation along the one dimension that matters on a perpetual venue: the directionality of forced flow.

The crucial word in Proposition 1 is the conditional. The reservation lean is optimal if liquidation flow is toxic — if a fill taken into the cascade is adversely selected. Whether that premise holds is an empirical question, and the rest of the paper answers it.

5. Data and empirical results

This section does four things. It describes the liquidation data and the cascade construction. It estimates the bivariate Hawkes process the model assumes. It then tests the model's central empirical implications on Hyperliquid. The tests do not support the model's premise on the most liquid contract. I report this directly: on Bitcoin, forced-liquidation flow is not the toxic adverse-selection channel the model treats it as. Liquidation cascades mean-revert, and passive liquidity provided into them earns, on average, favourable forward returns. I set out the evidence, report the heterogeneous picture across contracts honestly, and discuss what it implies for the model's scope.

5.1. Data and cascade construction

The liquidation series is built from Hyperliquid trade- and fill-level data, which records every fill on the venue in the public API format. A forced liquidation is a fill carrying a liquidation flag with the liquidated account, the mark price, and the method. This source is independent of how a liquidation is routed: it captures liquidations that execute on the book and those absorbed by the protocol vault alike. This matters because liquidations on the venue do not pass through a single identifiable account; an approach based on a liquidator address would miss the on-book majority. The signed direction of each liquidation follows from the liquidated account's position: a forced-sold long is a long liquidation, a forced-bought short is a short liquidation.

Raw liquidation fills cluster mechanically. A single liquidation order sweeps several price levels and prints as many fills sharing a timestamp. Treated as separate events these would inflate the self-excitation estimate. The inter-event gaps are sharply bimodal: a large mass at zero and within a few milliseconds, then a near-empty band below one second, then genuine gaps of seconds to minutes. I therefore collapse same-

direction liquidations separated by less than one second into a single cascade event, timed at onset, with a mark equal to the total liquidated notional. The one-second bandwidth sits in the empty band, so any choice between one millisecond and one second yields the same grouping; the construction is robust by inspection and is verified to be stable across all three contracts studied. For Bitcoin over 1–28 May 2026 this yields 1,087 long and 2,233 short cascades; the corresponding counts are 383 and 833 for Ether and 329 and 925 for Solana.

5.2. The liquidation cascade process

I fit the bivariate exponential-kernel Hawkes process of Section 2 to the cascade onsets by maximum likelihood, using the recursive exponential-kernel likelihood of Ogata (1981), which is linear in the number of events. Fit is assessed by the time-rescaling theorem (Brown et al. 2002; Daley and Vere-Jones 2003): under the estimated model the compensated inter-event times of each component are unit-rate exponential, which I check by a Kolmogorov–Smirnov test. A single stationary fit over the full month is rejected, for a reason the daily counts make plain: the process is non-stationary, with the dominant direction switching across market regimes. I therefore report three specifications. A whole-sample fit, a rolling daily fit, and a time-varying-baseline fit with one shared excitation kernel and a separate baseline per day; the last is the preferred form, as it encodes the finding directly: stable cascade mechanics with a regime-varying arrival rate.

The rolling fit is over the 19 of 28 days on which each side has at least eight cascades, the threshold below which the kernel parameters are not separately identified; the thinner days are omitted from the rolling estimation only. The preferred time-varying-baseline specification does not face this threshold: it uses the entire continuous month for the shared kernel, while the day-specific baselines absorb the quiet days natively, so no day is dropped. Across the rolling windows the structural parameters are stable while the baseline is not. The spectral radius is below one in every window, with median 0.81 and interquartile range [0.78, 0.85]; the decay corresponds to a fifteen-to-seventeen-second timescale across specifications. The preferred time-varying-baseline fit gives the estimates in Table 1.

Table 1. Bivariate Hawkes estimates, Bitcoin liquidation cascades, 1–28 May 2026. Normalised exponential kernel, so amplitudes are branching ratios.

Quantity	Whole-month	Rolling median [IQR]	TVB (preferred)
a_{++} long→long (self)	0.73	0.67 [0.54, 0.76]	0.730
a_{--} short→short (self)	0.62	0.46 [0.23, 0.61]	0.620
a_{+-} long→short (cross)	0.35	0.41 [0.35, 0.61]	0.350
a_{-+} short→long (cross)	0.04	0.15 [0.01, 0.23]	0.041
β decay (per second)	0.058	0.068 [0.060, 0.095]	0.063
$\rho(\Gamma)$ spectral radius	0.82	0.81 [0.78, 0.85]	0.81
decay timescale $1/\beta$	17.2 s	≈ 15 s	16.0 s

Note: $\log L = 17,699.9$ (whole-month, 7 params, AIC 35,413.7) versus $-17,571.4$ (TVB, 59 params, AIC 35,260.9). The TVB specification is preferred on penalised fit, lowering the Akaike criterion by about 153 despite 52 additional baseline parameters.

Three features are stable across every specification. The endogenous Hawkes clusters are subcritical, with spectral radius near 0.81, so each cascade cluster dies out almost surely, even though new cascades continue to arrive through the baseline intensity. Self-excitation dominates: a long-liquidation cascade generates an

expected 0.73 direct long-cascade offspring, a short cascade 0.62 direct short offspring. The cross-channel is strongly asymmetric: a long cascade generates an expected 0.35 direct short-cascade offspring, while a short cascade generates only 0.04 direct long offspring. The endogenous excitation decays over a fifteen-to-seventeen-second horizon. For comparison, Cao and Gang (2026) fit a multivariate Hawkes process to cross-protocol liquidations on Ethereum lending markets and report a subcritical regime with spectral radius 0.725 and self-excitation ratios of 0.28 to 0.32. Although the settings are not directly comparable — cross-protocol lending versus a single perpetual venue, with differing event construction — both regimes are subcritical, while the single-venue perpetual cascades exhibit markedly larger same-side excitation, consistent with a single matching engine concentrating feedback that is dispersed across protocols in the lending setting.

The cross-channel asymmetry is worth one observation, offered as interpretation rather than established mechanism. A long cascade exciting subsequent shorts far more than the reverse (0.35 against 0.04) is consistent with the mean-reversion documented below: forced selling overshoots, and the rebound that follows is what brings short positions under pressure, so long cascades seed short ones while the opposite channel stays weak. I note this consistency but do not claim the asymmetry causes the favourable adverse-selection result; the sample is also net short-liquidation-heavy, which the day-specific baselines absorb but which leaves the cross-channel only partially identified. The data permit this reading; they do not compel it. The same process structure recurs on Ether ($\rho = 0.77$) and Solana ($\rho = 0.82$), with the same self-excitation dominance and the same cross-channel asymmetry, so the cascade dynamics are not a Bitcoin artefact.

5.3. Tests of the model's implications

Proposition 1 rests on a premise: that forced-liquidation flow is the dominant source of adverse selection for a passive maker, so the maker should quote away from the active cascade direction. I test this premise and its consequences with three measurements on the venue's order book and trade tape, reported for Bitcoin, the most liquid contract and the one with the most cascades. I report all three, including those of opposite sign to the model. The reconstructed conditional intensity from the fitted kernel serves as the regressor throughout, so the tests do not depend on matching individual trades to liquidations. All standard errors are heteroskedasticity- and autocorrelation-consistent (HAC, 20 lags).

Adverse selection. If liquidation flow is toxic, a passive fill taken while same-side liquidation intensity is high should be followed by an adverse mark move. I classify each trade by the passive side (a marketable sell hitting the bid leaves the maker long; a marketable buy leaves the maker short), measure the signed forward mark move relative to subsequent marks at horizons from one to one hundred and twenty seconds, and regress this measure on the contemporaneous same-side excitation intensity. The premise predicts a positive coefficient. The estimate is negative and increasingly so with horizon: indistinguishable from zero at one to two seconds, then negative from five seconds, reaching about -0.0036 by sixty seconds and flat thereafter. Passive fills on the cascade side are not picked off; they earn favourable forward mark movement, consistent with profitability before costs, and the gain accrues over roughly the first minute. This is the opposite sign to the premise.

Depth. If the maker leans away from the cascade, aggregate resting depth should tilt away from the side facing elevated liquidation intensity. I regress the top-of-book depth imbalance on the excitation intensity gap, controlling for contemporaneous liquidation notional and return. The coefficient is positive at the top of book and significant: depth tilts toward, not away from, the heavier cascade side. Liquidity is supplied

into the cascade, not withdrawn from it. The effect weakens deeper in the book and is insignificant by twenty levels.

Spread. If cascade pressure raises the risk of providing liquidity, the half-spread should widen with total liquidation intensity. Regressing the relative half-spread on total excitation intensity, the raw association is positive but disappears once realized volatility is controlled. There is no liquidation-specific widening beyond the ordinary volatility channel.

Table 2. Panel of tests, Bitcoin, 1–28 May 2026. HAC(20) standard errors. ‘Opposite sign’ denotes a coefficient significant with sign opposite to the toxicity premise.

Test (premise prediction)	Coefficient	t	Verdict
Adverse selection, 5 s (positive)	−0.0010	−4.2	opposite sign
Adverse selection, 60 s (positive)	−0.0036	−5.0	opposite sign
Depth lean, top-1 (negative)	+2.31	+2.5	opposite sign
Depth lean, top-20 (negative)	+0.92	+1.5	null
Spread widening (positive)	≈ 0	+1.5	null

5.4. Interpretation: cascades mean-revert

The three tests are mutually consistent and point to a single mechanism. On Bitcoin a liquidation cascade is an overshoot, not informed flow. Forced selling pushes the price below subsequent marks; over the following minute it reverts. A passive bid filled during a long-liquidation cascade therefore buys at a transient discount and gains as the price recovers, which is the negative adverse-selection coefficient that builds over sixty seconds. Liquidity providers, recognising this, supply depth into the cascade rather than withdrawing it, which is the positive depth-imbalance coefficient. And because the flow is not toxic, there is no liquidation-specific spread premium beyond volatility. The horizon profile of the adverse-selection measure, near zero on impact and growing to a plateau, is the signature of mean reversion rather than of measurement error.

This is the opposite sign to the premise of Proposition 1. The model assumes liquidation flow is the dominant adverse-selection channel and derives that the maker should skew away from the active cascade. On Bitcoin the maker should, if anything, skew toward it. The model is internally correct; its empirical premise does not hold on this contract.

5.5. Heterogeneity across contracts and a depth hypothesis

The Bitcoin result is clean, but honesty requires reporting that it does not replicate uniformly across the venue's contracts, and that one contract points the other way. I ran the identical panel on Ether and Solana, with each contract's own fitted kernel.

On Ether the tests are null throughout: the adverse-selection coefficient leans negative at fifteen seconds but is not significant at any horizon, and neither depth nor spread shows a significant relation. This is most plausibly a power problem rather than a contradiction. Ether has fewer than half the cascades of Bitcoin, the excitation gap is positive only six percent of the time, and the adverse-selection regressor is an order of magnitude more dispersed; the test simply cannot detect an effect of the Bitcoin magnitude. A null here is consistent with the Bitcoin effect being present but undetectable.

On Solana the picture is mixed and, on two of three tests, leans toward the toxicity the model assumes. Adverse selection is favourable but only reaches significance at the sixty-second horizon. The depth coefficient, however, is negative and significant at the top of book — depth leans away from the cascade, the sign Proposition 1 predicts — and the half-spread widens with cascade intensity even after the volatility control. These are the opposite of the Bitcoin findings. I do not over-read them: Solana has the thinnest data of the three, the coefficients are marginal (p between 0.01 and 0.03), the depth effect vanishes by twenty levels, and across three contracts and three tests with multiple horizons some crossings of conventional significance are expected by chance. Solana is suggestive, not conclusive.

Taken together the three contracts say something more nuanced than a clean replication, and more interesting. The cascade process structure replicates across all three — subcritical, self-exciting, the same asymmetry. The economic toxicity, however, appears contract-specific, and it lines up with book depth. Average top-of-book-to-twenty notional depth over the sample is roughly fifteen million dollars on Ether, six million on Bitcoin, and six hundred thousand on Solana. The toxicity ordering matches the inverse: Solana, the thinnest book, is the one showing depth withdrawal and spread widening; Bitcoin and Ether, the deeper books, mean-revert or show nothing. This suggests a pre-registered hypothesis, which the present sample can motivate but not establish:

Depth-toxicity hypothesis. Liquidation toxicity scales inversely with order-book depth. A thin book cannot absorb forced flow, so a cascade moves the price and it stays moved — the toxic case the model is built for. A deep book absorbs the cascade, which then reverts — the favourable case observed on Bitcoin. The falsifiable prediction is a monotonic ordering: rank contracts by book depth, and the adverse-selection and depth-imbalance coefficients should move from toxic to favourable as depth rises.

This is a hypothesis raised by three points of data, two of which rest on marginal coefficients from the thinnest sample. I therefore flag it as the natural next test rather than a finding, and have begun collecting the data to adjudicate it: an extended Hyperliquid window, and a parallel liquidation series from a second venue spanning a wider range of book depths, against which the pre-registered ordering can be tested with adequate power. Establishing the depth-toxicity relation — and with it the precise venue and contract conditions under which the quoting result of Section 4 binds — is left to that work.

6. Conclusion

Forced liquidation is the defining microstructural feature of perpetual futures, and the natural conjecture is that it is the dominant source of adverse selection for a liquidity provider on these venues. I modelled signed liquidation flow as a bivariate self-exciting process and embedded its conditional intensity in an Avellaneda–Stoikov control problem. The maker who is filled and left with inventory must unwind it into a book consumed by same-side forced flow; I derived this exit cost rather than assuming it, and showed it is quadratic in inventory and linear in the matched intensity. That microfounded cost, and nothing else in the model, breaks the inventory symmetry of the value function. The consequence is sharp and conditional: if liquidation flow is toxic, then at flat inventory the optimal reservation price shifts away from the more active cascade direction, linearly in the intensity gap, while the half-spread carries no such first-order asymmetry. Classical and persistent-order-flow market making are recovered in the no-liquidation limit.

I then tested the premise the prediction rests on. Using a complete, routing-independent on-chain liquidation series for Hyperliquid, I characterised the cascade process — subcritical, self-exciting, directional, decaying

over tens of seconds — and tested whether the flow is toxic. On Bitcoin it is not: cascades mean-revert, passive fills into them earn favourable forward returns, depth is supplied into the cascade, and there is no liquidation-specific spread premium. The maker should lean into the cascade, not away. Across contracts the picture is heterogeneous — clean on Bitcoin, underpowered on Ether, mixed and leaning toxic on the thin-book Solana — which motivates a pre-registered hypothesis tying toxicity to book depth. The model is internally correct; its empirical premise is venue- and contract-specific, holding where backstop liquidity is thin and failing where it is deep. The contribution is a tractable model whose scope these results delimit, a reusable on-chain liquidation dataset and cascade construction, and direct evidence on when liquidation-aware quoting should matter. Testing the depth-toxicity hypothesis on venues spanning a wider range of book depth, and the maker-level test that aggregate book data cannot deliver, are the natural next steps.

References

- Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.
- Avellaneda, M. and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217–224.
- Bacry, E., Mastromatteo, I. and Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1), 1550005.
- Brown, E. N., Barbieri, R., Ventura, V., Kass, R. E. and Frank, L. M. (2002). The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2), 325–346.
- Cao, D. and Gang, P. (2026). DeFi liquidations cluster across protocols in a multivariate Hawkes framework. Working paper, SSRN 6508318.
- Cartea, Á., Jaimungal, S. and Penalva, J. (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press, Cambridge.
- Daley, D. J. and Vere-Jones, D. (2003). *An Introduction to the Theory of Point Processes, Volume I*, 2nd ed. Springer, New York.
- Guéant, O., Lehalle, C.-A. and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4), 477–507.
- Hawkes, A. G. (1971). Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1), 83–90.
- Ho, T. and Stoll, H. R. (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1), 47–73.
- Jusselin, P. (2021). Optimal market making with persistent order flow. *SIAM Journal on Financial Mathematics*, 12(3), 1150–1188.
- Ogata, Y. (1981). On Lewis’ simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1), 23–31.